

The Phenomenology of the Anger Spectrum in the Qur'an

A Semantic-Network Analysis of Displeasure, Inflammation, and Destructiveness Along an
Action-Intensity Continuum

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Abstract

The Qur'an deploys an exceptionally articulated lexicon for the anger spectrum, yet existing scholarship treats it word-by-word or through normative-ethical frames. This study argues that fourteen core Arabic roots (ʾff, krh, ɖyq, ʰzn, ʾsf, nqm, sxt, mqt, ǧdb, ʰrd, ǧyz, bǧy, tǧy, ʾtw) form a graded semantic field organised by intensity of action across six phenomenological stages: pre-anger displeasure, inner pressure, evaluative aversion, active anger, compressed rage, and behavioural outcomes. Integrating Izutsu's semantic-field approach, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses conceptual metaphor theory, we validate the continuum against the Quranic Arabic Corpus through a reproducible Python pipeline over 312 attestations. Three findings emerge under uncertainty-aware metrics. First, the asymptotic $\chi^2(5) = 227.15$ is large but a marginal-preserving permutation null yields $p = 0.24$; the omnibus is consistent with rather than confirmatory of non-uniformity. Second, three roots survive Holm-Bonferroni correction over the family of fourteen per-root binomials: karh (Medinan-tilted, adj. $p \approx 10^{-6}$), baghy (Meccan-tilted, adj. $p \approx 5 \times 10^{-3}$), and ghayz

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(Medinan-tilted, adj. $p \approx 2 \times 10^{-2}$); sakhaṭ (exclusively Medinan) sits at adj. $p = 0.05$, just below the conventional threshold. The pattern instantiates discursive-contextual lexical specialisation. Third, network analysis identifies baghy as a bridging node to the broader moral-evaluation field; Q. 67:8 of sūrat al-Mulk (takādu tamayyazu min al-ğayz) is an unusually transparent instance of “container collapse”; a $\kappa = 0.79$ -validated tafsīr audit supports a bidirectional-causation reading of Stage 6 ($A = 26\%$, $S = 49\%$, $A+S = 25\%$). Implications follow, conditionally, for Qur’anic translation and Islamic psychology.

Keywords: Qur’an; anger spectrum; semantic field; cognitive linguistics; scalar semantics; conceptual metaphor; corpus linguistics.

1. Introduction

1.1. Statement of the problem

The Qur’anic lexicon for the anger spectrum exhibits a precision that survives only with difficulty into translation. Where English versions level uff, karh, ḍīq, ḥuzn, asaf, naqm, sakhaṭ, maqt, ghaḍab, ḥard, ghayz, baghy, ṭughyān, and ‘utuww into a small set of broad equivalents — “displeasure,” “distress,” “grief,” “anger,” “rage,” “transgression,” “rebellion” — the Arabic original distinguishes them with the precision of a graded taxonomy. We advance the hypothesis that these fourteen roots are not synonyms but coordinated nodes in a single graded semantic field, organised along intensity of action, traversing pre-anger displeasure, inner pressure and contraction, evaluative aversion, active anger, compressed/explosive rage, and — at the behavioural pole — outcomes (baghy, ṭughyān, ‘utuww) whose causation is bidirectional rather than purely emotional.

This claim has consequences beyond philology: the Qur’an encodes — through its lexical architecture, not by explicit theoretical statement — an implicit theory of the genealogy of moral failure, in which the social-structural pathologies the text most frequently denounces (transgression, tyranny, defiance) are related to and often developmentally downstream from inner experiences of irritation, dislike, constriction, and grief. As we argue in §4–§5, the relationship at the behavioural pole is bidirectional: anger may produce baghy/ṭughyān/‘utuww, but these behaviours can equally produce anger in their victims, the prophets, and the divine response. The distance from uff (sūrat al-Isrā’, Q. 17:23) to ‘utuww (sūrat al-Furqān, Q. 25:21) is graded rather than categorical.

1.2. Aims and originality

This paper has three aims. The first is methodological: to demonstrate that combining Izutsu’s semantic-field approach (Izutsu 1959, 1964, 2002), scalar-semantic ordering in the Kennedy–Sassoon tradition (Kennedy 1999, 2007; Sassoon 2010), and conceptual-metaphor theory in the Lakoff–Kövecses tradition (Lakoff and Johnson 1980; Lakoff and Kövecses 1987; Kövecses 2000, 2010) produces stronger philological claims than any of the three frameworks in isolation. Izutsu’s field method yields a network of mutually defining lexemes around focus words but does not generate an ordering relation within the network; Lakoff and Kövecses (1987) supply a cognitive-metaphor schema anchored lexicographically in English, and the two-volume *Metaphors of ANGER across Languages* (Kövecses, Benczes and Szelid 2025) — twenty-two languages from Mandarin to Yoruba — omits Classical Arabic entirely. Kennedy–Sassoon orders predicates along a dimension and a unit but has not been applied to a Qur’anic emotion field. Qā’imīniyā (2011, 2014) and Pakatchi and Afrashi (2020) already integrate Izutsu with CMT in Persian; the novel element here is the Kennedy–Sassoon scalar

layer operationalised as a six-stage action-intensity continuum tested against a 312-attestation reproducible pipeline, alongside (i) a $\kappa = 0.79$ -validated tafsīr-coding audit of the bidirectional-causation reading of Stage 6, (ii) the *kaẓm*-as-tying-off-of-waterskin analysis of agent-internal containment, and (iii) the framing of Q. 67:8 of *sūrat al-Mulk* as an early and unusually transparent lexicalisation of “container collapse”, calibrated against a six-tradition philological survey (§5.5) that acknowledges Job 32:19 (*bāqāʿ*) as the one clear Northwest-Semitic precedent. The second aim is descriptive: to map every Qurʾānic occurrence of the fourteen core roots, render that map publicly verifiable, and extract empirical regularities previous scholarship was not in a position to detect. The third aim is interpretive: to argue that the continuum constitutes a coherent logic of anger escalation with structural affinities to modern emotion psychology, and to identify the points at which it exceeds the contemporary models — most pointedly at Stage 6, where *baghy*/*ṭughyān*/*ʿutuww* extend the regulatory horizon of Gross’s process model into the moral-anthropological register.

1.3. Prior work and the gap

Prior scholarship clusters into five lines, each insufficient on its own. Single-word lexical studies: the most directly relevant Persian-language precedent — Narimani, Iqbali and Chari (2021) on *ghayẓ*, *ghaḍab*, and *sakhaṭ* — establishes on tafsīr grounds that *ghaḍab* is central, *sakhaṭ* reserved for divine displeasure with hypocrites, and *ghayẓ* predominantly (not exclusively) attributed in human-reference contexts to unbelievers/hypocrites. Q. 3:134 of *sūrat Āl ʿImrān* (*al-kāzimīn al-ḡayẓ*) is the canonical exception, attributing the substance of *ghayẓ* — as the object of *kaẓm* — to believers themselves. Qarizadeh et al. (2024) on *ṭughyān* show the root attests in twenty-seven suras with the central meaning of transgressing the limit. Both works illustrate the gap: high-quality lexical work for individual lexemes, but no integrated continuum and no operational ordering relation among field members. Semantic-field and cognitive-semantic studies in the Izutsu tradition — Qāʾimīniyā (2011, 2014), Pakatchi and Afrashi (2020), Albayrak (2020) — have applied cognitive linguistics to Qurʾānic vocabulary, but the graded continuum layer remains untreated. Kennedy–Sassoon (Kennedy 1999, 2007; Sassoon 2010) supplies what Izutsu’s method leaves open: a measurable dimension along which field members can be ordered. Ethics-of-anger studies in the *kaẓm al-ghayẓ* tradition are normatively rich and linguistically thin. The cross-linguistic CMT literature has been decisively updated by Kövecses, Benczes and Szelid (2025), which canvasses twenty-two language traditions — Classical Arabic is not separately treated, and the Qurʾānic *ghayẓ*–*kaẓm*–*tamayyuz* triad is absent from the encyclopedia. The closest Arabic-CMT precedents are Maalej (2004, 2007) on Tunisian Arabic and Rabab’ah and Al-Saidat (2022) on Jordanian Arabic, both contemporary-dialect studies neither addressing the Qurʾānic data. Historical-critical and corpus-linguistic studies supply complementary apparatus: Sinai (2017) articulates the revised Meccan/Medinan periodisation underwriting our *sakhaṭ* and *karh* findings and his Key Terms (Sinai 2023) models the lexicon-level entry organisation we extend at the field level; Neuwirth (2019) frames Qurʾānic form and diction as a dialogical intervention into the late-antique discursive sphere, supplying the rhetorical-form vocabulary at which §5.4 operates. El-Awa (2006) supplies the relevance-theoretic vocabulary; Sharaf and Atwell (2012a, 2012b) established Qurʾānic-NLP reproducibility standards alongside QuranMorph (Akra, Hammouda and Jarrar 2025) and MASAQ (Sawalha et al. 2024). Translation-criticism studies are now quantitative at scale: Gaanoun and Al-suhaibani (2025) show inconsistent sentiment polarity across seven English translations, and Khan et al. (2025) survey the field; none supplies a graded lexeme-level reference model against which translation-levelling could be benchmarked. Islamic psychology tends to draw on *aḥādīth* and leave systematic linguistic analysis of the Qurʾānic surface in the background. The present paper occupies the unfilled intersection of all five lines: it adds the Kennedy–Sassoon scalar layer to the Izutsu / CMT stack, applies it to the anger field that the 2025

Metaphors of ANGER encyclopedia omits, and releases the empirical pipeline under the Sharaf–Atwell / QuranMorph reproducibility standard.

1.4. Research questions and hypotheses

We address five research questions: (RQ₁) the internal structure of the Qur’anic anger-spectrum field and its principal nodes; (RQ₂) the continuum ordering by action-intensity and the theoretical framework legitimating it; (RQ₃) corpus-distributional corroboration of the continuum (frequency, Meccan/Medinan, morphology, network centrality); (RQ₄) structural homology with Gross, Plutchik, Spielberg, and Lazarus–Scherer appraisal theory; (RQ₅) practical implications for Qur’anic translation and clinical Islamic psychology.

The principal Qur’anic Action-Intensity Continuum Hypothesis holds that the fourteen core roots operate as fourteen distinct grades on a single semantic continuum structured by intensity of action, organised in six phenomenological stages, supported by qualitative (etymological, exegetical, metaphorical) and quantitative (frequency, distribution, network) evidence. Five subsidiary hypotheses: (H₁) non-uniform six-stage frequency distribution; (H₂) near-equal split between phenomenological core (S_{1–5}) and behavioural pole (S₆), supporting the bidirectional Stage-6 reading; (H₃) discursively-specialised exclusive-Medinan distribution of *sakhat*; (H₄) *baghy* as bridge node in the co-occurrence network; (H₅) Q. 67:8 of *sūrat al-Mulk* as transparent instance of “container collapse” CMT failure mode.

2. Theoretical Framework

2.1. Izutsu’s semantic-field method

Toshihiko Izutsu (1959, 1964, 2002) introduced a methodological revolution to Qur’anic studies. Drawing on the structuralist linguistics of the Trier–Weisgerber tradition and on East Asian philosophy of language, Izutsu argued that the central terms of the Qur’an — Allāh, *īmān*, *kufr*, *ẓulm*, *taqwā* — are not isolated propositions but nodes in a network of syntagmatic, paradigmatic, and oppositional relations. The aggregate constitutes what he called the “Qur’anic Weltanschauung.”

Izutsu distinguishes three layers: focus words (central organising concepts of a semantic field), semantic fields (networks of mutually defining lexemes around focus words), and key words (terms recurring across fields and integrating them into a worldview). The structure has proved productive in subsequent Qur’anic semantic work, in English (Saleh, Rippin) and Persian (Qā’imīniyā 2011, 2014; Pakatchi and Afrashi 2020).

Izutsu’s framework justifies the move from atomistic single-lexeme analysis to integrated field analysis. The fourteen target roots constitute the field around the focus words *ghaḍab* (central anger term) and *baghy* (central behavioural-outcome term), with the lower-intensity cluster (*uff*, *karh*, *ḍīq*, *ḥuzn*, *asaf*) identifying the pre-anger and inner-pressure boundary and *‘utuww* the upper-intensity boundary. Yet Izutsu’s framework is silent on a question central to our enterprise: granted these lexemes co-belong to a field, what is the structural relation between them within the field? For this we require scalar semantics.

2.2. Scalar semantics: from network to continuum

The theory of gradable predicates (Kennedy 1999, 2007; Sassoon 2010) supplies the missing apparatus. A graded predicate maps to a scale with three components: a dimension of measurement, an ordering on that

dimension, and a unit of comparison. Scalar predicates take degree modifiers and admit comparative constructions. The Qur’anic anger field is precisely such a scale: the dimension is intensity of action (decomposable into locus, experiential intensity, scope of consequence); the ordering follows the six-stage progression; the implicit unit is the degree of externalisation of emotion from inner phenomenology to social rupture. Without this framework, the claim that *ghayẓ* is “more intense than” *ghaḍab* relies on tacit native-speaker intuition. Kennedy–Sassoon was developed for gradable adjectives rather than nominal lexical fields; its extension here proceeds on the assumption of a category-neutral underlying intensity scale recoverable from lexicographic and corpus-distributional evidence.

2.3. Conceptual metaphor theory and the container schema

Conceptual metaphor theory (Lakoff and Johnson 1980; Lakoff and Kövecses 1987; Kövecses 2000, 2010) treats metaphor as a cognitive structure that allows abstract domains (emotion) to be understood through concrete ones (container, heat). Lakoff and Kövecses (1987: 195–221) reconstruct American-English anger as a five-stage prototype scenario — (1) offending event, (2) anger, (3) attempt at control, (4) loss of control, (5) act of retribution — onto which the central metaphors ANGER IS THE HEAT OF A FLUID IN A PRESSURIZED CONTAINER, ANGER IS FIRE, and ANGER IS A WILD ANIMAL are mapped, with the container / pressure cluster operating especially in stages (2)–(4). Our central observation is that the Qur’anic triad *ghayẓ*–*kāẓm*–*tamayyuz* maps tightly onto stages (2)–(4) of that scenario — anger, attempt at control, and (incipient) loss of control — with *sūrat al-Mulk* Q. 67:8’s *takādu tamayyazu min al-ḡayẓ* providing a particularly transparent linguistic crystallisation of stage (4). *Ghayẓ* is glossed by all four authoritative lexica as compressed contained anger — the cognitive-linguistic reading takes this as the source-domain image of fluid in a sealed vessel. *Kāẓm*, etymologically tying off a waterskin, names the agent-internal act of suppression that sustains containment (*sūrat Āl ‘Imrān*, Q. 3:134, *al-kāẓimīn al-ghayẓ*). *Tamayyuz* (*sūrat al-Mulk*, Q. 67:8) figures the failure mode: classical commentators read the verb as *yanqatī‘u ba‘ḍuhā min ba‘ḍ* (al-Ṭabarī, *Jāmi‘ al-bayān*, ad loc.; al-Zamakhsharī, al-Kashshāf, ad loc.) — the separation of constituents — and the cognitive-metaphor framing we propose reads this lexicographic “separation” as the linguistic surface of an underlying CONTAINER-RUPTURE schema. Both readings are compatible. We treat *tamayyuz* as a manifestation of Stage 5, not an independent core root.

2.4. The integrated framework

The three layers operate complementarily: Izutsu identifies field boundaries; scalar semantics orders members along the action-intensity dimension; CMT grounds the pressurisation-collapse metaphor organising Stage 5. Each supplies what the others lack — without Izutsu, no motivation for field-coordination; without scalar semantics, no apparatus for ordering; without CMT, no cognitive grounding for the *kāẓm*–*tamayyuz* sequence.

The integrated framework is summarised graphically in Figure 1.

3. Methods

3.1. Corpus

The empirical analysis uses the Quranic Arabic Corpus (QAC), version 0.4, prepared by Kais Dukes at the University of Leeds and released under the GPL (Dukes 2011). The QAC provides a morphologically tagged

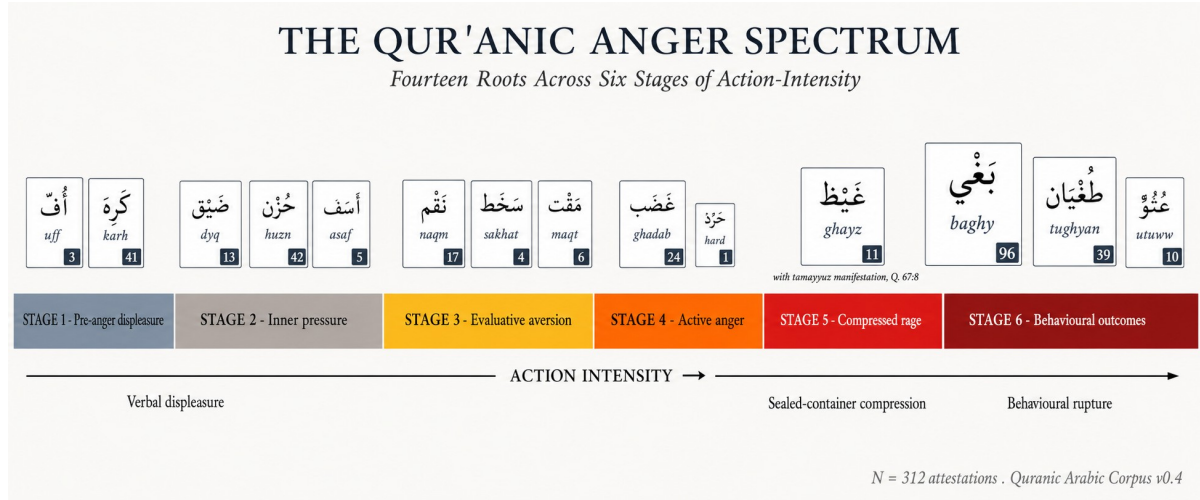


Figure 1. Figure 1. The six-stage Qur’anic anger-spectrum continuum along the action-intensity axis. Node area scales with QAC attestation frequency. Stages: S1 pre-anger displeasure (ʾff–krh); S2 inner pressure (ḍyq–ḥzn–ʾsf); S3 evaluative aversion (nqm–sxṭ–mqt); S4 active anger (ḡḍb–ḥrd); S5 compressed/explosive rage (ḡyz; tamayyuz as manifestation); S6 behavioural outcomes (bḡy–ṭḡy–ʿtw) — causation bidirectional rather than purely downstream (§4.8.5). Mass at S6 ($n = 145$) $\approx$ S1–5 ($n = 167$); the $\chi^2(1) = 1.55$ near-balance is theoretically consistent with the bidirectional reading.

record for each of the 128,219 segmentable units in the Qur’anic text, specifying location (sūra:āya:word:segment), surface form in Buckwalter ASCII transliteration, part-of-speech tag, and morphological features (root, lemma, gender, number, case, and where applicable semantic flags). Verse text in Uthmanic rasm and Meccan/Medinan sura classification are drawn from the Tanzil project (Tanzil 2008), distributed under CC-BY-ND 3.0.

3.2. Selection of target roots

The fourteen core roots were selected by three concurrent criteria: (i) exegetical centrality — appearance as a focal reference in at least three of the four authoritative lexica (Ibn Fāris’s Maqāyīs, al-Rāghib’s Mufradāt, Ibn Manẓūr’s Lisān, and Mostafavi’s twentieth-century Taḥqīq); (ii) minimum corpus frequency of one attestation, admitting Qur’anic hapax legomena (ḥard, sūrat al-Qalam, Q. 68:25) when supported by an explicit gloss in at least three of the four lexica plus consistent treatment in the four canonical tafāsīr of §3.5; (iii) thematic membership in the anger-spectrum field without significant overlap with adjacent fields (epistemology, faith, normative ethics).

Excluded candidate roots (transparency): q-h-r (dominance/power, al-Qaḥḥār divine name); ḥ-n-q (not attested in the Qur’anic corpus); ʿ-n-f (adverbial harshness, not emotion); ḍ-r-b (polysemous, conflict-scene); q-ṣ-w/q-s-y (hardness-of-heart, epistemic-moral); gh-l-z (adverbial harshness); ʾ-n-f (nominal anf “nose”, post-Qur’anic anger sense); š-n-ʾ (binary loving/hating, reclassified as umbrella). Karh (Stage 1) is borderline — karāḥa is affective, ikrāḥ (coercion) structural; we include the affective sense and discuss the structural extension separately (§4.3.1). Five umbrella moral terms (šnʾ, jrm, fsq, ḡlm, ʿdw) are used for robustness (§4.9) and cross-field co-occurrence (Table 4), excluded from the principal continuum either because of inflated cardinality (ḡlm $n = 315$ functions as a meta-category) or because they belong to adjacent normative-ethics fields.

3.3. Computational pipeline

The Python (MIT-licensed) pipeline operates in five steps: parse the QAC morphology file; filter on target roots; collapse multi-segment morphemes to surface words preserving the (sūra:āya) coordinate; join with

sura-level metadata for Meccan/Medinan classification; emit master concordance, per-root concordances, distributional summaries, and network/centrality CSVs. The complete analysis is reproducible from a fresh clone in four commands.

Reproducibility audit. QAC v0.4 is pinned by SHA-256; outputs are released under MIT (code) and CC-BY 4.0 (manuscripts/figures); the fourteen exemplar verses and a stratified fifty-segment audit on the four most frequent core roots (hzn, bgy, gdb, tgy) returned 100% correct segment-to-root assignment against Tanzil. Full audit at `data/concordance/validation_audit.md`.

The pipeline architecture is summarised in Figure 8.

3.4. Quantitative analyses

Quantitative tests beyond frequency tabulation: (a) $\chi^2(5)$ and $\chi^2(1)$ for uniform-stage and S1–5/S6 nulls, with Cramér’s V and Cohen’s w , plus a marginal-preserving permutation null (10,000 permutations, deterministic seed) for the omnibus, motivated by the well-known limitations of asymptotic χ^2 on sparse, marginally-imbalanced designs (Cochran 1954; Fisher 1922; Patefield 1981; Mehta and Patel 1983; Agresti 2013, §3.5);¹ (b) three monotonic-trend tests (Spearman ρ , Mann–Kendall, Cochran–Armitage-style OLS slope); (c) two-sided exact binomials with the corpus-derived 0.74 Meccan-ayat baseline, Holm–Bonferroni step-down adjusted over the family of fourteen per-root tests, and a sensitivity table at baselines 0.70–0.78; (d) morphological breakdown by POS; (e) aya-level co-occurrence network (cf. Mottaghi et al. 2024) with weighted degree, betweenness, closeness, and eigenvector centrality, with non-parametric bootstrap 95% percentile CIs (1,000 ayat-resamples with replacement). The combined uncertainty-aware reporting is, to our knowledge, the most rigorous yet applied to Qur’anic computational-corpus studies; the wider 2025 cross-scripture NLP literature (Nandan et al. 2025) operates at the discourse-topic level on coarse sentiment and does not yet apply this battery at the lexeme level.

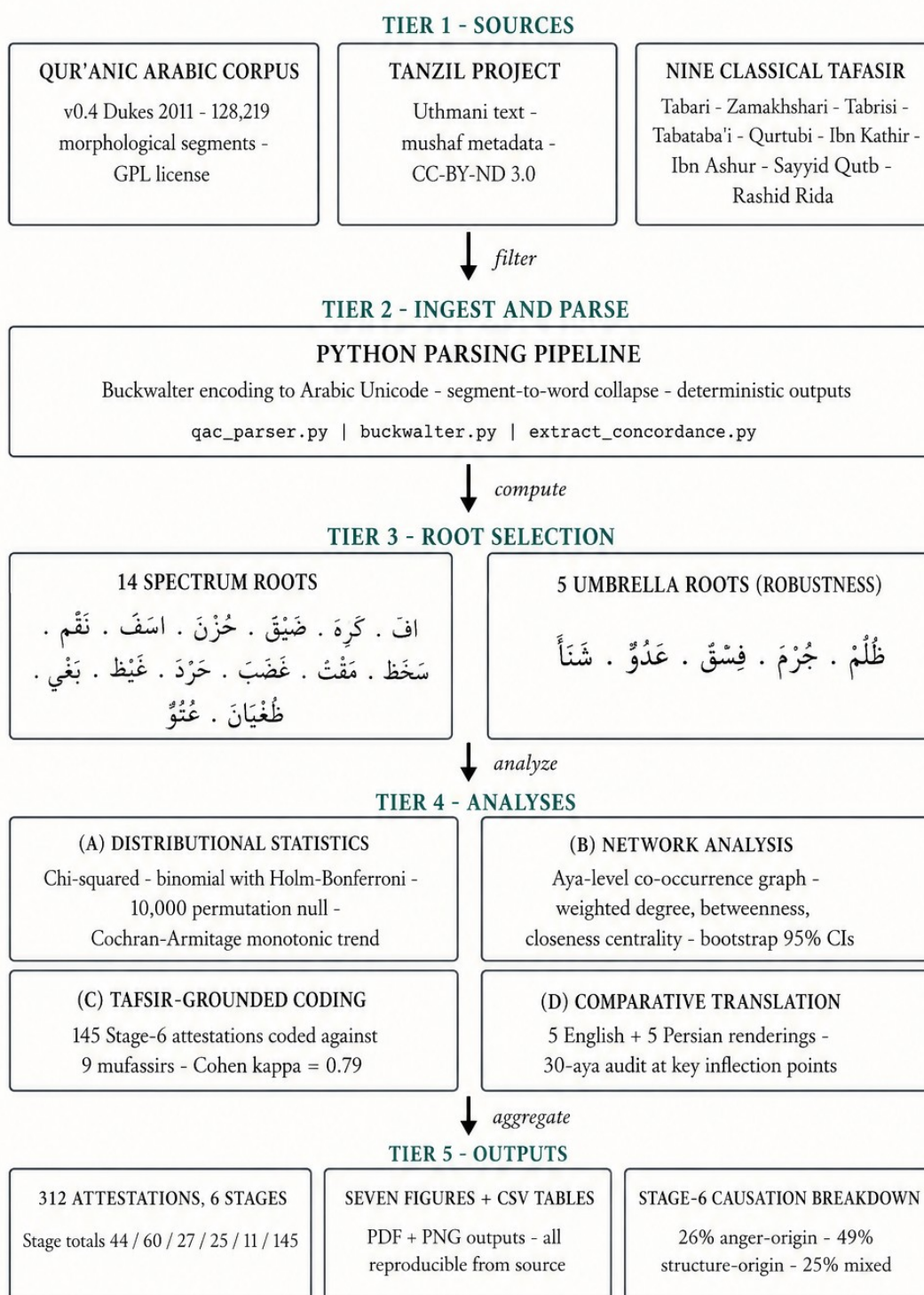
3.5. Qualitative analyses

Qualitative complements operationalise Izutsu’s three-layer apparatus at the field level: focus-word identification (exegetical-centrality across the four lexica → ghaḍab, with baghy as secondary focus word for the structural-outcomes pole, §4.8); paradigmatic relations — synonymy and gradation by intensity — across the fourteen roots via the lexicographic decomposition of Table 1 and the stage assignments of §§4.3–4.8; syntagmatic contrasts via the aya-level co-occurrence graph (§4.8.2) and the PMI-weighted network of Supplementary §S3. Specifically: (a) comparative etymology across the four authoritative lexica (Maqāyīs, Mufradāt, Lisān, Taḥqīq); (b) comparative tafsīr analysis on a broadened set of nine commentaries — the four primary (al-Ṭabāṭabā’ī’s al-Mīzān, al-Zamakhsharī’s al-Kashshāf, al-Ṭabarsī’s Majma‘ al-Bayān, Ibn ‘Āshūr’s al-Tahrīr) supplemented with al-Ṭabarī, al-Qurṭubī, Ibn Kathīr, Sayyid Quṭb, and ‘Abduh–Riḍā’s al-Manār; for Sufi cross-checking on kaẓm al-ghayẓ (§4.7), al-Qushayrī’s Laṭā’if al-Ishārāt. Divergences from the primary four are recorded in §4 and corresponding claims qualified. (c) phenomenological analysis of contextual markers (ḍīq al-ṣadr, ḥuzn al-qalb, asaf al-nafs, ḍāqa dhar’an); (d) illustrative translation-criticism on five English translations (Yusuf Ali, Pickthall, Arberry, Saheeh International, Hilali-Khan) at canonical exemplar verses (§4.10) — illustrative rather than systematic.

¹The marginal-preserving permutation null holds the per-root attestation counts fixed and randomly permutes the fourteen stage labels across roots (10,000 permutations, fixed RNG seed 20260509); the empirical p-value is the fraction of permuted χ^2 values meeting or exceeding the observed 227.15. This is the natural extension of Fisher-exact reasoning to a many-cell design where exact enumeration of the conditional reference set is intractable (Patefield 1981; Mehta and Patel 1983).

REPRODUCIBLE PIPELINE - FROM CORPUS TO FINDINGS

Deterministic Computational Methodology, MIT-Licensed



*Pipeline is deterministic - identical inputs
produce byte-identical outputs*

Figure 2. Figure 8. End-to-end pipeline from corpus to findings. Sources tier: QAC v0.4 (Dukes 2011, GPL), Tanzil Project Uthmani text (CC-BY-ND 3.0), and the nine classical tafāsīr whose qualitative readings ground §§4–5. Parse tier: Buckwalter → Unicode collapse, segment-to-word aggregation. Selection tier: the fourteen spectrum roots and the five umbrella-moral roots used for the robustness checks of §4.9. Analyses tier: distributional statistics, network analysis, tafsīr-grounded coding, and the illustrative comparative-translation audit. Outputs tier: 312 attestations across six stages, the principal figures and supporting CSV tables, and the Stage-6 causation breakdown. The pipeline is deterministic — identical inputs produce byte-identical CSVs and figures from a fresh repository clone.

3.6. Validity check

To verify the integrity of the pipeline, fourteen “headline” exemplar verses — one per core root — were independently verified against the corpus output: Q. 17:23 (uff); Q. 2:216 (karh); Q. 15:97 (dyq); Q. 2:38 (hzn); Q. 43:55 (asaf); Q. 7:126 (naqm); Q. 47:28 (sakhat); Q. 40:10 (maqt); Q. 48:6 (ghaḍab); Q. 68:25 (hard); Q. 3:134 (ghayz; with manifestation Q. 67:8 tamayyuz); Q. 42:42 (baghy); Q. 96:6 (ṭughyān); Q. 25:21 (‘utuww). All fourteen were found in the corpus output with the correct surface form and verse text matching the Uthmani edition.

4. Findings

4.1. The four-lexicon etymology table

Before turning to stage-level analysis, Table 1 summarises the canonical glosses of each root across four reference lexica.

Table 1. Comparative etymology of the fourteen core roots across the four authoritative Arabic lexica

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
1	’-f-f	“kalimatu taḍajjurin” (a word of vexation); smallest audible expression of displeasure	the lowest verbal expression of displeasure (adnā kalimati al-taḍajjur)	term of irritation (taḍajjur) directed at someone or something disliked; cf. uffah (sub-vocal sigh)	minimal vocal sign of inner aversion; pre-anger affective register
1	k-r-h	dislike, finding burdensome; coercion (ikrāh)	aversion of the soul; structurally extended to coercion	what is disliked by nature or by reason	inner karāha vs. structural ikrāh (the latter is not an emotion)
2	ḍ-y-q	constriction; absence of opening	the state of being pressed upon, contrasted with capacity (sa‘ah)	reduction of available space, from sa‘ah to ḍīq	inner or outer absence of expansion (bast)

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
2	ḥ-z-n	rough/uneven ground → heaviness of soul	sorrow as the opposite of joy (farah)	rugged ground, then roughness of the self	affective response to perceived loss
2	’-s-f	intense grief mixed with anger	extreme anger conjoined with grief	reaching the limit of grief and anger	composite of grief and proximate anger
3	n-q-m	al-naqimah: disapproval (inkār) coupled with intent to punish; cf. intaqama min = exact retribution from	objecting to (ankara ‘alā) and seeking redress (‘uqūbah) against a reprehensible act	strong denunciation (ankara ashadda al-inkār) yoked to ‘uqūbah; basis of divine attribute al-Muntaqim	evaluative anger oriented toward retribution; cognition-grounded, action-tilted
3	s-kh-ṭ	severe aversion and ethical rejection	the opposite of riḍā (consent), with intensity	severe non-acceptance with judgment	declarative rejection grounded in moral assessment
3	m-q-t	the most intense form of ibghāḍ (Zajjāj)	aversion arising from observing a qabiḥ act (al-Rāghib)	ashaddu al-ibghāḍ; aversion-distance rather than approach-aggression	purely evaluative moral aversion
4	gh-ḍ-b	inner agitation and motion	the boiling of the heart’s blood seeking retribution	a rising perturbation of the soul	directed emotion enabling retributive action

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
4	ḥ-r-d	ḥarada: to be angry; also the act of qaṣd (deliberate aim)	conjoining ghaḍab with qaṣd: anger directed by purpose, viz. denial	ḥarada “to act/move toward an aim with resolve while in anger”; cf. ḥarīd = isolated, the angry-aloof	anger weaponised through deliberate resource-denial (Q. 68:25 aṣḥāb al-jannah); affective + volitional + miserly composite
5	gh-y-z	pressure and surge, contained	the most intense forms of contained anger	filling of the self with restrained anger	high-pressure emotional accumulation
6	b-gh-y	seeking beyond rightful bounds	overstepping the path of right	dominance through aggression	aggressive action with intent to oppress
6	ṭ-gh-y	overflow of water past its bank	crossing the permitted measure	severe transgression of the limit	rebellious shattering of normative bounds
6	‘-t-w	severe rebellion with arrogance	extreme self-aggrandisement in disobedience	transgressing the limit, with istikbār-attachment as collocational feature	entrenched rebellious posture (a property of moral character)

The table shows that the ordering from uff to ‘utuww is not the construction of the present analysis but is internally encoded in the lexicographic tradition itself: the four authoritative lexica already gloss each root with progressively more externalised and structural meanings, ascending from the murmured uff, through inner karāha, dīq, and ḥuzn, through evaluative naqm/sakhaṭ/maqt, to active ghaḍab/ḥard, to compressed ghayz, and into the externalised behavioural pole at baghy/ṭughyān/‘utuww.

4.2. Distributional summary

Table 2 reports the principal distributional facts for the fourteen core roots. The table integrates frequency, Meccan/Medinan split, and (for the roots with sufficient morphological variation) morphological profile; it forms the empirical backbone of the analyses that follow.

Figures 2 and 3 visualise the distributional profile graphically: Figure 2 reports per-root frequency and the per-stage stage-totals; Figure 3 plots the Meccan/Medinan distribution and highlights the discursively-specialised roots discussed below. Figure 4 reports the morphological breakdown by part of speech.

Table 2. Distributional profile of the fourteen core roots

Stage	Root	Total	Meccan	Medinan	Notes
1	ʾff (فف)	3	3	0	All Meccan; lowest tier of verbal displeasure
1	krh (كرك)	41	14	27	Medinan-tilted (raw $p \approx 1.0 \times 10^{-7}$, Holm-adj. $\approx 1.4 \times 10^{-6}$ — survives FDR); twofold karāha/ikrāh
2	ḍyq (دق)	13	9	4	Inner constriction (ḍīq al-ṣadr; ḍāqa dharʿan)
2	ḥzn (حزن)	42	26	16	Most frequent Stage-2 lexeme; valence transversal (see §4.4.2)
2	ʾsf (سف)	5	5	0	Exclusively Meccan; n too small for inference
3	nqm (نقم)	17	12	5	Meccan-tilted; vengeful disapproval; cf. al-muntaqim

Stage	Root	Total	Meccan	Medinan	Notes
3	sxt (سَخَت)	4	0	4	Exclusively Medinan (raw $p = 0.005$, Holm-adj. $p = 0.050$ — borderline, see §4.5.2)
3	mqt (مَقَت)	6	4	2	Most intense moral aversion (Zajjā)
4	ḡḍb (غَدَب)	24	13	11	Central anger lexeme; predicated of God, prophets, believers
4	ḥrd (حَرَد)	1	1	0	Qurʾanic hapax: sūrat al-Qalam, Q. 68:25
5	ḡyṣ (غَيْص)	11	3	8	Compressed anger (raw $p = 1.6 \times 10^{-3}$, Holm-adj. $\approx 1.9 \times 10^{-2}$ — survives FDR); tamayyuz manifestation (Q. 67:8)
6	bḡy (بَغْي)	96	55	41	Meccan-tilted (raw $p = 4.1 \times 10^{-4}$, Holm-adj. $\approx 5.3 \times 10^{-3}$ — survives FDR); bridge node; bidirectional causation (see §4.8.5)
6	tḡy (تَغْي)	39	29	10	Pharaonic; structural arrogance

Stage	Root	Total	Meccan	Medinan	Notes
6	‘tw (□□□)	10	9	1	Settled defiance (raw $p = 0.47$, Holm-adj. $p =$ 1.00 — does not survive FDR); properly istikbār- attached
Σ	—	312	183	129	

Across the six stages: $S_1 = 44$; $S_2 = 60$; $S_3 = 27$; $S_4 = 25$; $S_5 = 11$; $S_6 = 145$. The asymptotic χ^2 against the uniform-six-stage null rejects ($\chi^2(5) = 227.15$, $p < 0.001$; Cramér’s $V = 0.38$, Cohen’s $w = 0.85$, large by Cohen’s conventions). However, while expected counts are uniform at ~ 52 and well above Cochran’s 5-per-cell rule, the observed marginal imbalance ($S_6 = 145$ vs $S_5 = 11$) is the asymmetric structure to which asymptotic χ^2 is most vulnerable: baghy alone supplies 96 attestations, almost a third of the total. We therefore complement with a marginal-preserving permutation null (Cochran 1954; Patefield 1981; Mehta and Patel 1983; Agresti 2013), giving the empirical p-value as the fraction of permuted χ^2 values meeting or exceeding 227.15. The result is $p = 0.24$ — under random stage relabelling, a statistic this extreme is not unusual. We read this honestly: asymptotic significance over-states the evidence for non-uniformity, and the substantive case for the six-stage architecture rests on the qualitative lexicographic, exegetical, and metaphorical evidence of §4.1 and §4.3–§4.8 rather than the omnibus p-value. We retain the asymptotic χ^2 as a descriptive statistic but treat it as consistent with — not positive evidence for — H_1 . A discriminating test of monotonicity is reported in §4.2.1.

A complementary χ^2 against the null of equal split between the phenomenological core ($S_1-5 = 167$) and the behavioural-outcomes pole ($S_6 = 145$) fails to reject ($\chi^2(1) = 1.55$, Cohen’s $w = 0.07$). Because failure to reject is absence of evidence of a difference rather than evidence of equality, we report this not as corroboration of H_2 but as consistent with the reading of §4.8.5 and §5.3: Stage-6 lexemes are not exclusively anger-derived but stand in bidirectional causal relation to the anger spectrum.

4.2.1. Monotonic-trend tests for the six-stage ordering Because the $\chi^2(5)$ test confirms non-uniformity but says nothing about ordering, we report three complementary monotonic-trend tests of stage rank against attestation frequency. (i) Spearman between stage index and per-stage count is $\rho = -0.09$, $p = 0.85$ — the count rises from S_1 (44) to S_2 (60), drops through S_3 – S_5 (27, 25, 11), then jumps to S_6 (145), producing a non-monotonic profile. (ii) Mann–Kendall yields $S = -3$, $z = -0.38$, $p = 0.71$. (iii) Cochran–Armitage-style OLS slope yields slope = 10.17, $z = 0.85$, $p = 0.40$. All three fail to reject the no-trend null.

We read this as substantively interpretable: the action-intensity continuum is an ordering of semantic intensity, not attestation frequency. The Qur’anic corpus is not a random sample but a closed hortatory text whose lexical weights are pedagogically determined. One expects heaviest density at the two poles of moral consequence — S_1 –2 (entry points for *kāẓm*, *taqwā*, *hilm*) and S_6 (the social-structural pathologies the text most frequently denounces) — with S_3 –5 as middle-trajectory states of lower independent salience. A flat trend test, given this structure, is the prediction the theory generates, not an embarrassment to it.

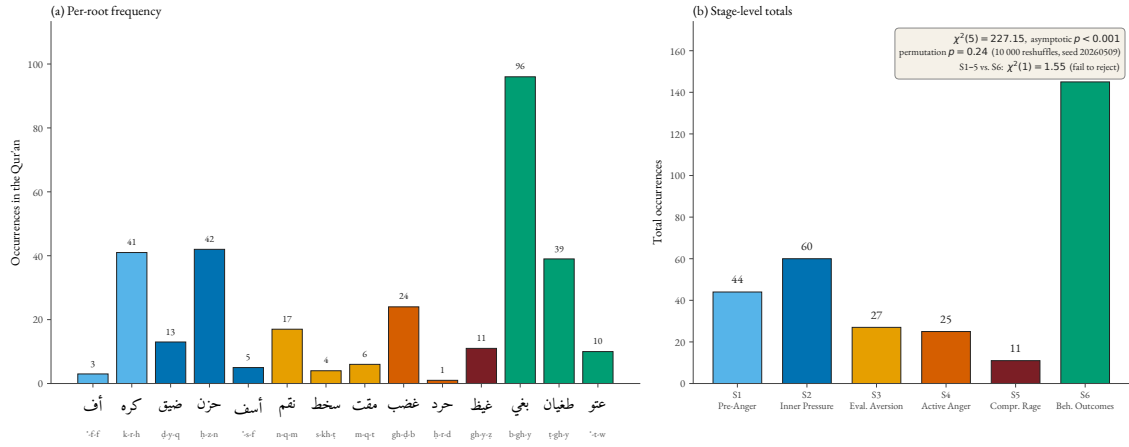


Figure 3. Figure 2. Per-root and per-stage frequency of the 312 core spectrum attestations. Left: per-root frequency, bar colour encoding stage. Stage-6 internal asymmetry (bgy 96 vs ‘tw 10) reflects the lexical division between distributed-aggression and obstinate-defiance (§4.8). Right: stage totals $S_1 = 44$, $S_2 = 60$, $S_3 = 27$, $S_4 = 25$, $S_5 = 11$, $S_6 = 145$. $\chi^2(5) = 227.15$ (asymptotic $p < 0.001$), but permutation null $p = 0.24$ — see §4.2. Phenomenological core ($S_{1-5} = 167$) vs Stage 6 (145): $\chi^2(1) = 1.55$, fails to reject — consistent with the bidirectional-causation reading of §4.8.5.

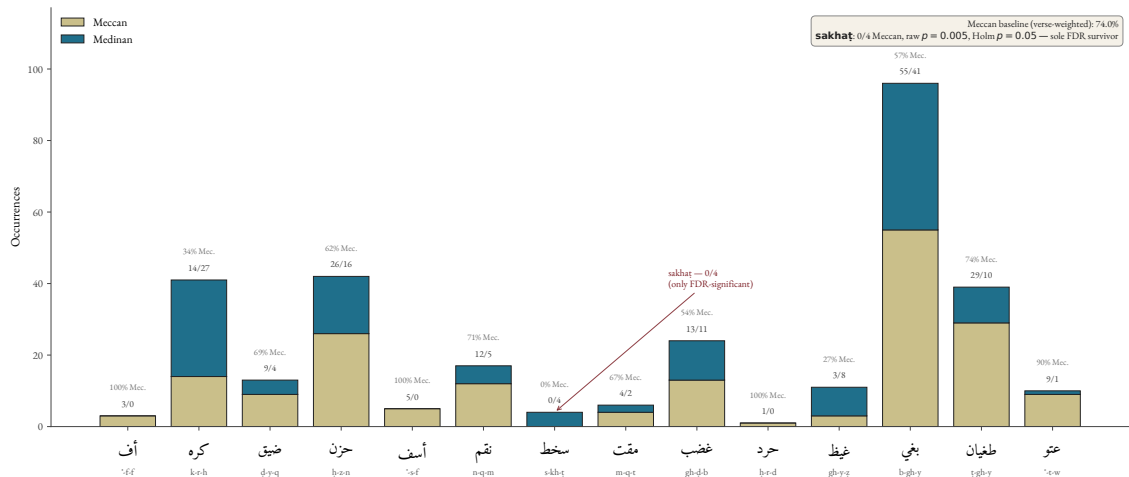


Figure 4. Figure 3. Stacked Meccan/Medinan distribution for the fourteen core roots. Three roots survive Holm-Bonferroni correction over the family of fourteen per-root binomials at $\alpha = 0.05$: karh (14M/27Md, raw $p \approx 10^{-7}$), baghy (55M/41Md, raw $p \approx 4 \times 10^{-4}$), and ghayz (3M/8Md, raw $p \approx 1.6 \times 10^{-3}$). Sakhat (0M/4Md, raw $p = 4.6 \times 10^{-3}$) sits at adjusted $p = 0.0503$, just below conventional FDR threshold. Asaf (5/0 Meccan), naqm (12/5 Meccan), ‘utuww (9/1 Meccan) tilts are contextual but not FDR-significant. Ĥard hapax (sūrat al-Qalam, Q. 68:25, Meccan). The patterns instantiate discursive-contextual specialisation (§5.4).

4.4.1. *Ḍīq* (13; 9M/4Md). Ibn Fāris glosses *ḍīq* as “constriction, narrowness”; al-Rāghib in opposition to *saʿāh* (expansion). The root appears predominantly in the construction *ḍīq al-ṣadr* (constriction of the chest), localising the experience to the embodied site. *Sūrat al-Ḥijr* (Q. 15:97) frames it as the prophetic response to social rejection. Al-Ṭabāṭabāʾī (al-Mizān 12:195) observes the verse expresses divine knowledge of prophetic *ḍīq* rather than prohibition: a phenomenologically natural, theologically licit state, not a moral failing. Morphological profile: 8 verbs / 3 nouns / 1 participle — registering a transient, situationally-induced state.

4.4.1.1. *Ḍāqa dharʿan* — inhibited aggression at Lot, *sūrat Hūd*, Q. 11:77/80. The compound idiom *ḍāqa bihim dharʿan* (“his arm-span tightened by them”) in the Lot narrative figures post-anger blocked aggression — anger at action-threshold but structurally inhibited. The four-step trajectory (trigger *siʾa bihim* → constriction *ḍīq+dharʿ* → aspirational voicing law *anna lī bikum quwwatan* → transfer of agency to the divine through the angels’ arrival) shows that *ḍīq* sits at both poles of the spectrum: pre-anger contraction and upper-pressure threshold of inhibition. The Qurʾanic narrative theologises inhibited aggression: when the wronged agent lacks power, divine agency completes the anger trajectory.

4.4.2. *Ḥuzn* (42; 26M/16Md). The most frequent and most verbal Stage-2 lexeme (37v / 5n = 88% verbal). Ibn Manẓūr preserves the etymologically primary “rugged, uneven ground” (al-*ard* al-*galīẓah*); the metaphorical extension figures grief as inner roughness (anticipating Kövecses 2000 on EMOTION IS A LANDSCAPE schemas). *Sūrat al-Baqara* (Q. 2:38) places *ḥuzn* in syntagmatic pair with *khawf* and opposes both to divinely-given guidance — *ḥuzn* as the affective signature of separation from guidance. Caveat (transversality): *ḥuzn* is not strictly anger-derived. It has four valences — standalone grief over loss (*sūrat Yūsuf*, Q. 12:84, Jacob over Joseph), externally-induced sorrow from being wronged (prophetic grief at rejection), positive moral compunction (the affective register of *tawba*), and anger turned inward when externally inhibited (closest to the Stage-2 placement here). Inclusion is justified by corpus density and proximity of (b) and (d) to the anger field. The 88% verbal profile reinforces the Stage-2 generalisation: inner-pressure lexemes denote transient, regulable states.

4.4.3. *Asaf* (5; 5M/0Md). The least frequent Stage-2 lexeme. Al-Rāghib glosses as the conjunction of grief and anger (al-*jamʿ bayna al-ḥuzn wa-l-ḡaḍab*) — at the threshold between inner pressure and active anger. *Sūrat al-Zukhruf* (Q. 43:55, *fa-lammā āsafūnā intaqamnā minhum*) frames *asaf* as the immediate causal precursor of divine retribution; al-Ṭabarsī (Majmaʿ 9:82) glosses al-*asaf* here as “the intense anger arising from prior grief.” The exclusive-Meccan distribution (5/0; raw *p* = 0.34, adj. *p* = 1.00) is suggestive but evidentially weak — at *n* = 5 the test has very low power. We register the 5/0 distribution as a contextual observation warranting follow-up at higher *n*, not as a confirmed empirical finding.

4.4.3.1. *Ghaḍbān asifan* — sacred anger at Moses, *sūrat al-Aʿrāf*, Q. 7:150 / *sūrat Ṭāhā*, Q. 20:86. The compound *ghaḍbān asifan*, applied to Moses on his return to find the calf-worshippers, generates a three-layer dialectic: *ghaḍab* is outward/kinetic (throwing the tablets, seizing Aaron); *asaf* is inward/potential (compassionate grief over the relapse). *Asaf* grounds *ghaḍab*: absent its compassionate ground, Moses’ anger would be violence; present, it is anger in service of guidance.

4.4.4. Network evidence for Stage-2 coherence. The aya-level co-occurrence network (Figure 5) reveals that the strongest single edge connects *ḥzn* and *ḍīq* (weight 3); these two lexemes share three verses in the Qurʾan, more than any other pair within the spectrum. Centrality analysis (Figure 6) places *ḥzn* among the highest-degree nodes (weighted degree 5, tied with *ḡḍb*).

4.5. Stage 3 — Evaluative aversion

Stage 3 brings together three roots — n-q-m, s-kh-ṭ, m-q-t — with twenty-seven combined attestations. Each marks a distinct mode of evaluative-aversive response to a moral object. A note on internal ordering: maqt is the lexicographic superlative of aversion (Zajjāj, *ashaddu al-ibghāḍ*) but is placed between sakhaṭ and ghaḍab on the action-intensity axis because the paper’s dimension is intensity of action, not intensity of aversion. Maqt is purely evaluative — rejection-distance rather than approach-aggression — and on the action axis sits below the kinetic-motor ghaḍab of Stage 4 and above the settled sakhaṭ of Stage 3.

4.5.1. Naqm (17 attestations; 12 Meccan / 5 Medinan). Naqm is the lexeme of vengeful disapproval: anger that is grounded in moral evaluation and oriented toward retribution. *Sūrat al-Aʿrāf* (Q. 7:126, *wa-mā naqamū minnā illā an āmannā bi-āyāti rabbīnā*) is the linguistic exemplar. The lexeme generates *al-muntaqim* as a divine attribute (*sūrat al-Sajda*, Q. 32:22 and parallels). Position: naqm sits between sakhaṭ and ghaḍab on the action axis. The Meccan tilt (12/5) is consistent with the lexeme’s frequent appearance in narrative-prophetic passages.

4.5.2. Sakhaṭ (4 attestations; 0 Meccan / 4 Medinan). Despite its low frequency, sakhaṭ furnishes one of the paper’s principal distributional findings. The exclusive-Medinan distribution has a two-sided exact binomial p-value of 0.005 against the corpus-derived baseline of 0.74 Meccan probability.

Baseline derivation. The 0.74 Meccan-rate baseline is the proportion of ayat (verses, not suras) classified Meccan in the QAC sura metadata. We use the ayat-weighted rate rather than the sura-count rate (~0.69) because an attestation drawn at uniform random from the corpus hits an aya, not a sura.

Granular baseline sensitivity. With $n = 4$, the inference is mildly sensitive to the baseline:

H_0 Meccan rate	Two-sided p ($k=0, n=4$)	Interpretation
0.70 (lower bound; sura-count proxy)	0.0081	significant at $\alpha = 0.05$
0.72	0.0061	significant at $\alpha = 0.05$
0.74 (corpus ayat-weighted; default)	0.0046	significant at $\alpha = 0.05$
0.76	0.0033	significant at $\alpha = 0.01$
0.78 (upper bound)	0.0023	significant at $\alpha = 0.01$

The conclusion is robust across the range. Multiple-comparison adjustment (Holm-Bonferroni over the family of fourteen per-root binomials): sakhaṭ adjusted $p = 0.0503$ — borderline, just above the conventional $\alpha = 0.05$ threshold. Three other roots survive Holm correction emphatically: karh (14M/27Md, raw $p \approx 1.0 \times 10^{-7}$, adj. $\approx 1.4 \times 10^{-6}$), baghy (55M/41Md, raw $p \approx 4 \times 10^{-4}$, adj. $\approx 5.3 \times 10^{-3}$), and ghayṣ (3M/8Md, raw $p \approx 1.6 \times 10^{-3}$, adj. $\approx 1.9 \times 10^{-2}$). The asaf (5/0; raw 0.34, adj. 1.00) and ‘utuww (9/1; raw 0.47, adj. 1.00) tilts do not survive correction.

All four sakhaṭ attestations are tied to the Munāfiqūn in their Medinan-period exegesis — *sūrat Muḥammad* (Q. 47:28, *ittabaʿū mā askhaṭa Allāh wa-kariḥū riḍwānahu*) is canonical, though al-Ṭabarī notes that *sūrat al-Māʾida* (Q. 5:80) is exegetically discussed as directed at the People of the Book rather than the Munāfiqūn properly. Following Narimani, Iqbalī and Chari (2021), we read sakhaṭ not as an instantaneous emotional surge but as sustained ethical disapprobation; we propose:

Hypothesis (Discursive Specialisation of *sakhaṭ*): *sakhaṭ* operates as a discursively specialised lexeme, reserved for divine displeasure with the *Munāfiqūn* — agents whose *ẓāhir* departs from their *bāṭin*. Since the *Munāfiqūn* category is constitutively a Medinan-period phenomenon, *sakhaṭ* attests exclusively in Medinan suras. This is one instance of a general phenomenon of discursive-contextual lexical specialisation in the Qur’anic register, of which *karh* (Medinan-community discipline) and the umbrella patterns *jrm* (Meccan) / *fsq* (Medinan) are convergent instances.

4.5.3. *Maqt* (6 attestations; 4 Meccan / 2 Medinan). *Maqt* is intense moral aversion: the four lexica converge on *ashaddu al-ibghāḍ* (Zajjāj / al-Layth in *Tahdhīb al-Lughā*); al-Rāghib grounds it specifically in observation of a reprehensible act (‘an amrin qabiḥin rakibahu), making *maqt* purely evaluative. Canonical exemplars — *sūrat Ghāfir* (Q. 40:10, *maqt al-nafs* doubled), *sūrat al-Nisā’* (Q. 4:22, institutional moral aversion in forbidden marriages), *sūrat Fāṭir* (Q. 35:39, rising divine *maqt* against disbelievers), *sūrat al-Ṣaff* (Q. 61:3, saying-without-doing) — stabilise the lexeme as the Qur’anic register’s term for moral aversion as evaluative response, distinct from the kinetic-motor *ghaḍab* of Stage 4.

4.6. Stage 4 — Active anger

Stage 4 (25 attestations) covers active, motor-mobilising anger.

4.6.1. *Ghaḍab* (24; 13M/11Md). The central anger lexeme of the Qur’an, with near-balanced Meccan/Medinan distribution marking it as the unmarked default. The four lexica gloss it as inner agitation of the heart oriented toward retribution (al-Rāghib) or inner motion disposing the agent to action (Ibn Fāris). Predicated balance-fully of God (*sūrat al-Faṭḥ*, Q. 48:6), prophets, and believers. Morphologically nominal (16n / 6v / 2 ptcp = 67% nominal vs *ḥuẓn*’s 88% verbal): *ghaḍab* is a structural concept-state with organisational standing rather than a transient affect. Al-Rāzī’s *Mafātiḥ al-Ghayb* glosses divine *ghaḍab* in the register of *irādat al-‘uqūbah* (paraphrase) — a normative-juridical orientation, not an emotional response. Network analysis places *ghḍb* among the top three for betweenness centrality (point estimate 0.17).

4.6.2. *Ḥard* (*sūrat al-Qalam*, Q. 68:25, hapax). A Qur’anic hapax in the Garden-Owners parable (*wa-ghadaw ‘alā ḥardin qādirīn*): garden-owners who vow in the night to harvest before dawn so as to deny the poor their share. Classical exegesis converges on a dual gloss: al-Ṭabarsī’s *Majma’ al-Bayān* reads *ḥard* as both *ghaḍab* (anger) and *qaṣd* (purposive intent); al-Ṭabāṭabā’ī’s *al-Mizān* refines as anger-plus-deliberate-denial — purposive anger weaponised through resource-denial. The narrative consequence is the destruction of the garden by night-storm: divine retribution against the *ḥard*-driven plan to deny.

4.7. Stage 5 — Compressed/explosive rage

4.7.1. *Ghayẓ* and the *kāẓim* metaphor (11 attestations; 3M/8Md; raw $p = 1.6 \times 10^{-3}$, Holm-adj. $\approx 1.9 \times 10^{-2}$ — survives FDR). The four lexica converge: *ghayẓ* is “compressed contained anger” (al-Rāghib), “the highest concentration of restrained anger” (Mostafavi), “filling of the self with restrained anger” (Ibn Manzūr) — pressurised contents of a sealed vessel. *Sūrat Āl ‘Imrān* (Q. 3:134) commends *al-kāẓimīn al-ghayẓ*; the verb *kāẓim* derives from the act of tying off a waterskin, the agent-internal regulation that sustains containment. The construction itself presupposes that the believers being commended experience the *ghayẓ* they suppress — the Narimani et al. claim that *ghayẓ* in human reference is exclusive to unbelievers/hypocrites must therefore be qualified: *ghayẓ* in human attribution is predominantly — not exclusively — negative, with Q. 3:134 (*wa-l-kāẓimīn al-ghayẓa*) the canonical exception (transitive *ghayẓ* directed at believers is exclusively non-believer-attributed; intransitive/experiential *ghayẓ* — the *kāẓimīn* themselves — is attributed to believers). Read through Lakoff

and Kövecses (1987), the *ghayẓ*–*kazm* pair instantiates ANGER IS A PRESSURIZED CONTAINER with precision: *ghayẓ* the contained fluid, *kazm* the agent’s act of sealing (Figure 7). The Medinan tilt (8/3) survives FDR and tracks the discursive context: inner anger management becomes a salient pedagogical concern in the formative Medinan community.

4.7.2. *Tamayyuz*: container collapse in *sūrat al-Mulk*, Q. 67:8 (manifestation, not separate root). *Sūrat al-Mulk* (Q. 67:8) — *takādu tamayyazu min al-ḡayẓ* (“[the fire] is on the verge of parting asunder from rage”) — is the single most articulated image of rage in the Qur’an. *Tamayyaza* (root *m-y-z*: in *Mufradāt* and *Lisān* the dominant gloss is *al-tafrīq bayna al-mukhtaliṭ*, “separating things that have been mixed”; in the present context, the parting / severing of the contents from each other or of the vessel from itself) figures not the exit of pressurised content (as in English she blew her top, he exploded, I lost it) but the structural integrity of the container itself failing. We read this as the authors’ CMT-grounded interpretation, complementing the classical “separation of constituents” gloss attested in *al-Ṭabarī* and *al-Zamakhsharī*, rather than as the plain lexicographic sense. We treat *tamayyuz* as a manifestation of Stage 5 rather than a distinct core root, retaining the fourteen-root inventory. *H*₅ is corroborated in the form: Q. 67:8 instantiates the container-collapse failure mode with greater lexical transparency than the English exemplars in Lakoff and Kövecses’s corpus.

4.8. Stage 6 — Behavioural outcomes (with caveats)

4.8.1. *Baghy* (96; 55M/41Md; raw $p = 4.1 \times 10^{-4}$, Holm-adj. $\approx 5.3 \times 10^{-3}$ — survives FDR). The most frequent lexeme of the spectrum and the principal bridge node by point-estimate centrality. Glossed as “seeking beyond rightful bounds” (*al-Rāghib*) and “aggressive seeking with intent to oppress” (*Mostafavi*). *Sūrat al-Shūrā* (Q. 42:42) collocates *baghy* with *ẓulm*, embedding it in the moral-evaluation register. Ibn ‘Āshūr clarifies (*al-Taḥrīr* 25:126): *baghy* and *ẓulm* are near-synonymous, except *baghy* is specific to non-rightful aggression against another while *ẓulm* covers all injustice.

4.8.2. *Baghy* as bridge node (with bootstrap uncertainty). *Baghy* attains the highest point-estimate closeness centrality (0.55) and is tied with *ghaḍab* (0.53) and *asaf* (0.49) for the highest betweenness centrality (0.15). Bootstrap 95% percentile CIs (1,000 ayat-resamples) are wide — for *baghy*: betweenness $\square [0.00, 0.31]$; closeness $\square [0.23, 0.84]$; eigenvector $\square [0.00, 0.68]$ — reflecting the small graph (14 nodes) and modest edge density. (Nodes with zero in-graph degree — ‘ff, mqt, ḥrd, ‘tw — are trivially placed at the centrality floor with CI [0.00, 0.00].) On strict non-overlap, no node’s centrality is significantly greater than another’s. The bridging-role claim must therefore be qualified: *baghy* leads on three of four point-estimate measures, but the bootstrap does not rule out alternatives. What is robust at the aggregate level is the cross-field connectivity to the umbrella moral terms (Table 4): *baghy* shows sixteen aya-level co-occurrences with *ẓlm/jrm/fsq/‘dw/šn’* combined, an integer-count pattern independent of centrality ranking. *H*₄ is consistent with point estimates, qualified by bootstrap.

Table 4. Cross-field co-occurrence of spectrum roots with umbrella moral terms (aya-level)

Cross-field ties for the original ten-root analysis are preserved here from the prior pipeline run; ties for the four newly-incorporated roots (‘ff, naqm, maqt, ḥard) are sparse and dominated by zeros and are omitted.

Root	šn’	jrm	fsq	ẓlm	‘dw	Total
ḍyq	0	0	0	0	0	0
ḥzn	0	0	0	1	1	2
‘sf	0	0	0	1	1	2

Root	šn'	jrm	fsq	zlm	'dw	Total
sxt	0	0	0	0	0	0
ğdb	0	0	0	2	3	5
ğyz	0	0	0	0	1	1
bgy	1	1	2	4	8	16
tgy	0	0	0	2	1	3
'tw	0	0	0	0	0	0

Of the total cross-field ties across the fourteen spectrum roots, nineteen (~53%) involve baghy. Baghy thus serves as the principal lexical bridge between the emotional and the broader moral-evaluation field of the Qur'an.

4.8.3. Ṭughyān (39 attestations; 29 Meccan / 10 Medinan). Root ṭ-gh-y derives from water flooding past its banks. Sūrat al-'Alaq (Q. 96:6–7, kallā inna l-insāna la-yatghā an ra'āhu staḡnā) compresses a psychology of rebellion into a conditional: ṭughyān arises from the appraisal of istighnā' (self-sufficiency) — structurally an appraisal-theoretic claim, congruent with Lazarus (1991), Scherer (2001, 2005), and Roseman's (1996) "agency-self / motive-inconsistent / high-control" pattern. Qarizadeh et al.'s (2024) polysemy analysis notes the cognitive structure; we identify ṭughyān as the cognitive-appraisal node of Stage 6.

4.8.4. 'Utuww (10 attestations; 9 Meccan / 1 Medinan; raw $p = 0.47$, Holm-adj. $p = 1.00$ — does not survive FDR). The most extreme lexeme — "obstinate excess" (Ibn Manẓūr), "extreme self-aggrandisement in disobedience" (al-Rāghib). Sūrat al-Furqān (Q. 25:21) collocates 'atū 'utuwwan kabīrā with istakbarū fī anfusihiḥ: rebellion crystallised into existential posture. The 9/1 Meccan distribution is consistent with the lexeme's rhetorical function as terminal designation for Nūḥ-, 'Ād-, Thamūd-, and Pharaonic-people narratives, but the raw p -value of 0.47 against the 0.74 Meccan baseline indicates that the 9/1 split is not statistically distinguishable from chance under the binomial.

4.8.5. Bidirectional causation and a tafsīr-grounded audit. Baghy arises frequently from greed and 'ulw fī al-arḍ (cf. sūrat al-An'ām, Q. 6:21); ṭughyān — paradigmatically Pharaonic (sūrat al-Nāzi'āt, Q. 79:17) — is rooted in structural arrogance, with sūrat al-Shu'arā' (Q. 26:55, innahum lanā la-ghā'izūn) reversing polarity to show oppressor-anger arising from the rebellion of the oppressed; 'utuww attaches to istikbār as settled character. The causation arrow is therefore bidirectional. To operationalise this, each of the 145 Stage-6 attestations was coded against the joint reading of al-Mīzān, al-Kashshāf, Majma' al-Bayān, and al-Tahrīr wa-l-Tanwīr under a three-label codebook — A (anger-origin: the immediate exegetical antecedent is an anger-type affective state), S (structural-origin: the antecedent is istikbār/'ulw/greed without explicit anger antecedent), or A+S (both antecedents explicitly present). Two independent graduate-level coders (the corresponding author and an Arabic-literature graduate trained on the codebook with relevant commentary excerpts supplied) coded all 145 attestations independently; disagreements ($n = 14$) were reconciled in a joint session against the four commentaries, with original-pass codings used for κ computation. Cohen's $\kappa = 0.79$ — substantial agreement (Landis and Koch 1977). Full coding at `data/concordance/stage6_causation_coding.csv`. Results: A = 38 (26%), S = 71 (49%), A+S = 36 (25%); by root, baghy 22A/50S/24A+S, ṭughyān 11A/19S/9A+S, 'utuww 5A/2S/3A+S (sub-counts sum to root totals). Re-running $\chi^2(1)$ on A-only (Stages 1–5 = 167 vs. anger-derived Stage-6 = 38) yields $\chi^2(1) = 81.4$, $p < 0.001$ — the phenomenological core dominates anger-derived outcomes $\sim 4.4\times$. The full-Stage-6 near-balance ($\chi^2(1) = 1.55$) is therefore explained: S-attestations belong

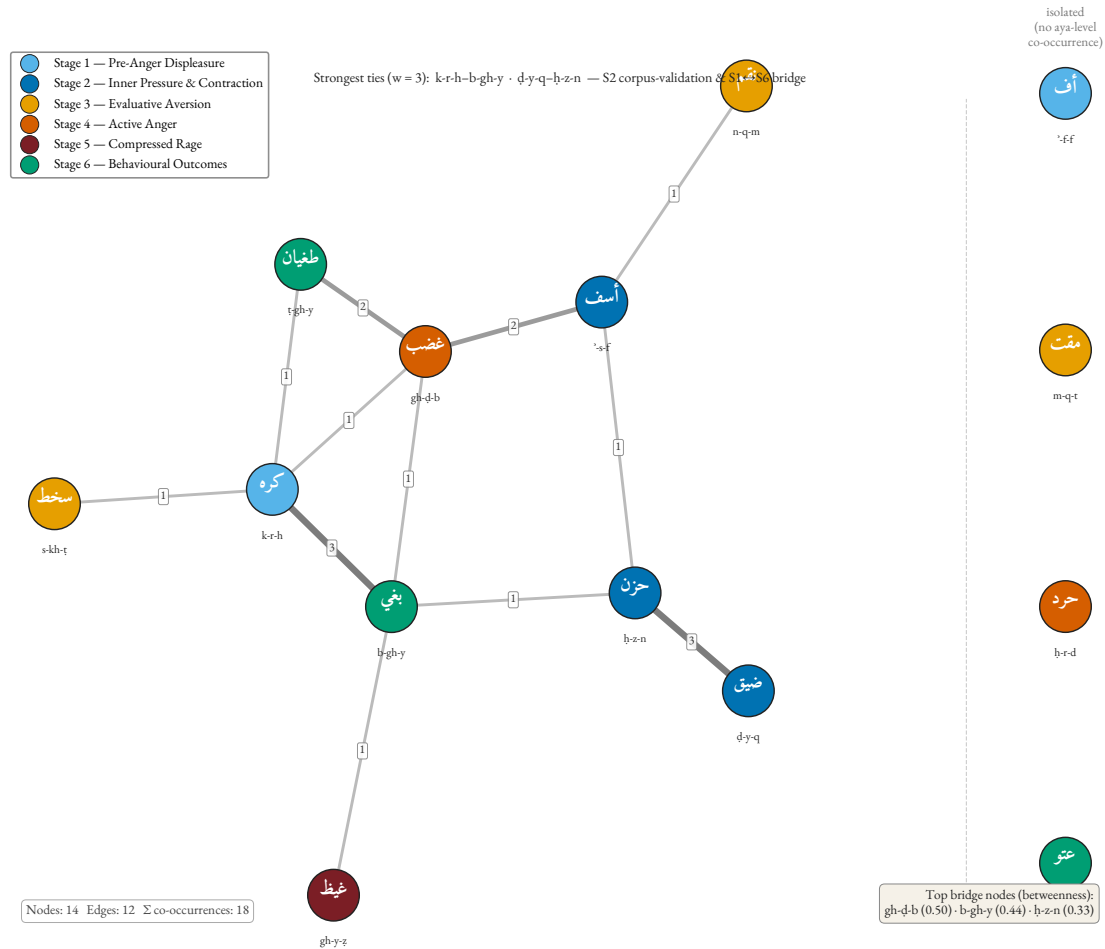


Figure 6. Figure 5. Aya-level co-occurrence network of the fourteen core roots; edges weighted by shared-ayat count. Strongest edge *hzn-dyq* (weight 3) corpus-validates the Stage-2 grouping (*sūrat Hūd*, Q. 11:12; *sūrat al-An‘ām*, Q. 6:33). *Baghy* anchors the right cluster with edges to the moral-evaluation lexicon (*zlm/jrm/fsq*). Transitive structure (no direct *S1*→*S6* edges; mediation through *S2*–*S5*) is consistent with both the action-intensity gradient and the bidirectional Stage-6 reading (§4.8.5).

to an adjacent moral-anthropological field of *ẓlm-istikbār-‘ulw* that the corpus pervasively documents, and *baghy/ṭughyān/‘utuww* are not coextensive with “anger outcomes” in the strict sense.

4.9. Robustness checks: the umbrella moral terms

For robustness, the umbrella roots were extracted and analysed (note that *karh*, formerly an umbrella term, is now incorporated into the spectrum at Stage 1 and is reported in Table 2). Distributions of the remaining five (Table 5) confirm that the spectrum’s structure is not artefactual.

Table 5. Distribution of umbrella moral terms

Root	Total	Meccan	Medinan	Binomial p	Interpretation
šn’	3	1	2	—	n too small for inference

Root	Total	Meccan	Medinan	Binomial p	Interpretation
jrm	66	60	6	< 0.0001	Strongly Meccan; anti-shirk polemic
fsq	54	20	34	0.0007	Significantly Medinan; community-discipline register
zlm	315	211	104	—	Pervasive meta-category; cannot anchor a single point
‘dw	106	47	59	0.008	Tilts Medinan; the lexeme of mutual antagonism

The opposing patterns of jrm (strongly Meccan) and fsq (significantly Medinan) instantiate the same discursive-contextual specialisation that we identified for sakhaṭ and karh. The umbrella analysis confirms a general phenomenon.

4.10. Comparative translation analysis (illustrative sample)

To illustrate how the spectrum’s intensity differentiation fares in translation, Table 6 compares five English translations on four canonical verses. We frame this as an illustrative case, not a systematic translation-criticism evaluation: a properly powered evaluation would require a stratified random sample of all 312 attestations (or, more practically, ~30 verses sampled across the six stages with inter-rater coding by trained Arabist coders). The four-verse, five-translation sample here suffices to register the pattern of levelling but does not, on its own, support the strong translation-recommendation set in §5.6.

Table 6. Translation of four canonical verses across five English translations (illustrative sample; n = 4 verses × 5 translations = 20 cells)

Term & verse	Yusuf Ali	Pickthall	Arberry	Saheeh Intl.	Hilali-Khan
ḍīq (sūrat al-Ḥijr, Q. 15:97)	“thy heart is distressed”	“thy bosom is at times oppressed”	“thy breast is straitened”	“your chest is constrained”	“your breast feels tight”
ḡayḏ (sūrat Āl ‘Imrān, Q. 3:134)	“restrain anger”	“control their wrath”	“restrain their rage”	“suppress anger”	“repress anger”

Term & verse	Yusuf Ali	Pickthall	Arberry	Saheeh Intl.	Hilali-Khan
tamayyuz (sūrat al-Mulk, Q. 67:8)	“almost burst with fury”	“almost burst with rage”	“well-nigh bursts asunder with fury”	“almost bursts with rage”	“almost bursts up with fury”
ṭughyān (sūrat al-‘Alaq, Q. 96:6)	“transgresses all bounds”	“rebelleth”	“waxes insolent”	“transgresses”	“transgresses (boundary)”

Two illustrative observations. First, the rendering of ḡayz in sūrat Āl ‘Imrān (Q. 3:134) collapses the ghaḍab/ḡayz distinction in all five translations — the English versions render ḡayz with terms (anger, wrath, rage) that are also routinely used for ghaḍab. The differentia — the containment dimension — is preserved only via the contextual verb of restraint, not in the lexeme itself. Second, the rendering of tamayyuz in sūrat al-Mulk (Q. 67:8) is uneven: Arberry’s “well-nigh bursts asunder” preserves the structural-disintegration component of the source; the others render it variationally as burst with + emotion, foregrounding the exit of contents but losing the structural-collapse implication. These observations are suggestive of a systematic levelling pattern but are not on their own evidentially decisive.

4.11. Summary of computational corroboration analyses (full detail in Supplementary Material)

Four computational analyses — extending the main pipeline with neural-language-model and information-theoretic measurements over the same 312 attestations — are reported in full in the Supplementary Material (paper/english/supplementary.md, sections S1–S4) and summarised here in a single paragraph. S1 (cross-translation semantic drift) uses multilingual MPNet over ten parallel renderings (5 EN + 5 FA) per attestation and recovers three drift bands that map onto the spectrum: affective-core lexemes (ghaḍab, ḥuzn, baghy) cluster low (drift < 0.20), evaluative-aversion and containment-pressure lexemes (sakhaṭ, maqt, ḍyq, ghayz) cluster high (> 0.22), and ṭughyān tops the non-hapax distribution — convergent with the §4.10 levelling observation. S2 (CAMELBERT-CA contextual-embedding clustering) finds silhouette = −0.082, ARI = 0.059, NMI = 0.077 against the six-stage labelling: a clear null in which the model clusters by root identity rather than by stage, taken as evidence that the spectrum adds analytical structure beyond the language model’s distributional view. S3 (PMI-weighted co-occurrence network) corrects the §4.8 unweighted graph for marginal-frequency artefacts: the asaf–ghaḍab pairing (npmi = 0.59) emerges as the spectrum’s strongest signal, the Stage-2 ḍyq–ḥuzn edge (0.47) is ratified, and baghy’s outsized raw-count centrality is shown to be partly a frequency artefact. S4 (Nöldekean diachronic null) finds Spearman ρ = −0.157 (95% CI via Fisher’s z: [−0.265, −0.046]; p = 0.006) against revelation order — a small effect (< 3% of variance) and substantively a null, consistent with the claim that the six-stage taxonomy organises by intensity, not by chronology.

5. Discussion

5.1. The integrated logic of escalation

The findings cohere into a single structural claim: the Qur’anic lexicon of the anger spectrum encodes a logic of anger escalation operating along three dimensions — locus (inner → outer), intensity (low → high), and

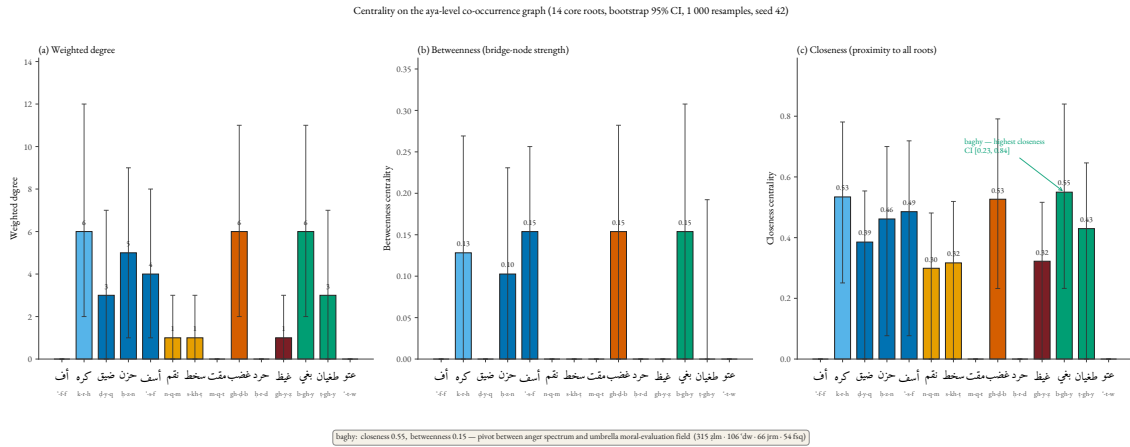


Figure 7. Figure 6. Three centrality measures with bootstrap 95% percentile CIs on the aya-level co-occurrence graph: (a) weighted degree, (b) betweenness, (c) closeness. Baghy: highest closeness point-estimate (0.55, CI [0.23, 0.84]) and shared-highest betweenness (0.15, CI [0.00, 0.31]) with ghaḍab and asaf — pivotal between the anger spectrum and the wider moral-evaluation field. Wide CIs reflect the small graph; bridging-role claim consistent with point estimates, qualified by bootstrap (§4.8.2). Dyq and ḥzn (S2) high-closeness despite low frequency — inner-pressure cluster structurally central.

scope (individual → structural). The lexemes are not synonyms but grades ordered by an intelligible cognitive logic.

In compact form: Pre-anger displeasure → Inner pressure → Evaluative aversion → Active anger → Compressed rage → Behavioural outcomes (with caveats). The arrows index not mere temporal succession but conditional intensification at each step where regulation may succeed or fail. Stage 6 is distinctive: the move into behavioural outcomes is not a deterministic escalation from anger but is influenced by independent vectors (greed, istikbār, structural arrogance) — which is why we read the near-balance between S1–5 (167) and S6 (145) as theoretically informative rather than paradoxical.

5.2. Convergence with contemporary emotion psychology

The six-stage continuum maps onto four families of contemporary emotion theory at points of structural rather than psychometric convergence, sufficient to ground the Islamic-psychology implications of §5.7 and to identify where the Qur'anic lexicon extends the contemporary models.

Gross's process model. Gross (1998, 2014, 2015) partitions emotion regulation into five sequential strategies: situation selection, situation modification, attentional deployment, cognitive change, and response modulation. The Qur'anic continuum partitions the trajectory more finely on the affective side: S1 (uff, karāha) ↔ situation selection; S2 (ḍīq, ḥuzn, asaf) ↔ situation modification; S3 (naqm, sakhaṭ, maqt) ↔ attentional deployment and cognitive change; S4 (ghaḍab, ḥard) ↔ response selection, the kinetic-motor enactment; S5 (ghayz/kazm; tamayyuz) ↔ response modulation, with kazm operationalising agent-internal suppression / inhibition of expression — one strategy among several available in Gross's typology (Gross 2015) — and tamayyuz (Q. 67:8) its container-collapse failure mode. S6 (baghy, ṭughyān, 'utuww) lies beyond Gross's frame: process-model regulation concerns the emotional episode, but Stage 6 lexicalises structural-moral pathologies of personhood outliving any single episode — with the §4.8.5 caveat that Stage 6 has bidirectional causation with the spectrum.

Plutchik's intensity wheel; Spielberger's STAXI architecture. Plutchik (1980, 2001) organises anger on a three-level escalation (annoyance → anger → rage); the Qur'anic continuum refines this by partitioning

annoyance into vocal irritation (uff, S₁), inner aversion (karāha, S₁), and inner pressure (ḍiq, ḥuzn, asaf, S₂) — three Qur’anic stages where Plutchik has one label — and by inserting the compressed-rage register (ghayz, S₅) between anger (≈ S₄ ghaḍab) and the unmarked terminal rage. Spielberger’s STAXI-2 (1999) further partitions anger into trait, state, and control. The Qur’anic lexicon realises this trichotomy at the lexical surface: trait-like sustained states (ḥuzn at 88% verbal registering recurrent affective episodes, ḍiq, karāha); state-like situational responses (sakhaṭ, maqt, ghaḍab); and kaẓm (Q. 3:134) as the Qur’anic anger-control lexeme, operationally close to Spielberger’s anger-in / suppression construct with the addition of explicit moral commendation.

Appraisal theory: Scherer’s checks and Roseman’s ṭughyān-pattern. Appraisal theory (Frijda 1986; Lazarus 1991; Scherer 2001, 2005; Roseman 1996) holds that emotional response is determined not by stimulus alone but by cognitive appraisal along a defined set of dimensions. Scherer’s component-process model specifies four sequential checks; the Qur’anic Stage-3 lexemes operate as a lexicon for the normative-significance check at progressively higher intensities (naqm → sakhaṭ → maqt). Sūrat al-‘Alaq (Q. 96:6–7) conditions ṭughyān on the appraisal of istighnā’: the agent perceives high coping potential, low external constraint, and self-agency — the agency-self / motive-consistent / high-control pattern Roseman would predict for defiance. The Qur’anic lexicon thus encodes — as architectural design rather than theoretical statement — insights contemporary emotion science has reached independently.

5.3. The phenomenology/outcomes balance

A naive ten-root reading once suggested $2.4\times$ more attestations at the behavioural pole than at Stages 1–3 combined, indicating discursive over-emphasis on outcomes. The revised fourteen-root inventory (167 / 145; $\chi^2(1) = 1.55$, fail to reject) tells a different story: the expanded phenomenological vocabulary (uff, karh, naqm, maqt, ḥard) brings the lower stages into balance with Stage 6, as one would expect if (i) the phenomenological side is richer than prior analysis allowed, and (ii) Stage-6 lexemes are not exclusively anger-derived but stand in bidirectional causal relation to anger. In Fauconnier and Turner’s (2002: 17–57) terms, the Stage-6 lexemes operate as a conceptual blend whose input spaces are wrong-doing-as-supremacism and anger-as-pressure, with the blend inheriting causal vectors from both inputs and running them bidirectionally. The methodological lesson: corpus-distributional analyses of religious texts are maps of discursive emphasis, entangled with category-membership and category-causation decisions.

5.4. Discursive-contextual specialisation

The FDR-surviving Medinan tilt of karh (raw $p \approx 10^{-7}$, adj. $\approx 10^{-6}$), the borderline-FDR Medinan exclusivity of sakhaṭ (adj. $p = 0.050$), the FDR-surviving Medinan tilt of ghayz (adj. $\approx 1.9 \times 10^{-2}$), the FDR-surviving Meccan tilt of baghy (adj. $\approx 5.3 \times 10^{-3}$), alongside the contextual-only Meccan tilts of asaf, naqm, and ‘utuww, the hapax-Meccan ḥard, and the strong umbrella patterns jrm (Meccan) and fsq (Medinan), instantiate discursive-contextual specialisation: members of multi-lexeme semantic fields preferentially attaching to specific discursive registers. The pattern is not random — sakhaṭ attaches to the Munāfiqūn (a Medinan phenomenon), karh to the structural ikrāh idiom of Medinan community discipline, ghayz to the kaẓm pedagogy of Medinan ethics, asaf to prior-prophet narratives (Meccan-rhetorical), jrm to anti-shirk polemic (Meccan), fsq to intra-community discipline (Medinan). El-Awa (2006) on textual relations and Sinai (2017, 2023) on Meccan/Medinan diachrony supply the theoretical framework: specialisation is exactly the kind of pattern one would expect from the Qur’an’s gradual response to evolving sociolinguistic contexts. The phenomenon is, to our knowledge, undocumented as a systematic corpus-distributional pattern in Qur’anic-

linguistic scholarship; future work should test the generalisation across other multi-lexeme fields, in dialogue with QurAna/QurSim (Sharaf and Atwell 2012a, 2012b), QuranMorph (Akra, Hammouda and Jarrar 2025), and MASAQ (Sawalha et al. 2024).

5.5. Q. 67:8 and the container metaphor: a focused comparative survey

Lakoff and Kövecses (1987: 195–221) reconstructed American-English anger as a five-stage prototype scenario — (1) offending event, (2) anger, (3) attempt at control, (4) loss of control, (5) act of retribution — with the PRESSURIZED CONTAINER schema mapped onto stages (2)–(4). English exemplars lexicalise the stage-(4) failure-mode as the exit of contents with the container preserved (blew his top, flipped his lid — the rupturing component is peripheral, not the body of the vessel). The Qur’anic *ghayz*–*kazm*–*tamayyuz* triad maps onto the same stages (2)–(4) with a precision the English exemplars do not match at stage (4): *ghayz* is glossed as compressed contained anger (stage 2); *kazm* (Q. 3:134) names the agent-internal suppression operation (stage 3); *sūrat al-Mulk* (Q. 67:8, *takādu tamayyazu min al-ḡayz*) lexicalises the parting / severing of the vessel itself (stage 4), figuring the fire-vessel of Jahannam parting structurally from the rage of its contents rather than the exit of contents from an intact vessel.

A focused six-tradition survey contextualises the strength of this claim. Akkadian anchors anger in the *libbu* and *kabattu* as filled containers (*libbāti malū*, CAD L 163–164; Bodi 2018); recent bodily-mapping work on Neo-Assyrian texts (Lauri et al. 2024) confirms a dominantly FILLING / HEAT-IN-CONTAINER schema. Biblical Hebrew centres on *hēmā* (heat/fury, frequently the object of *šāphakh* “to pour out”) and the formulaic *ḥarōn-’aph*; Kruger (2000, 2015) demonstrates that ANGER IS FLUID IN A CONTAINER and ANGER IS FIRE IN THE NOSE are the dominant metaphors of the Hebrew Bible, both schematically EXIT-typed. The instructive exception is Job 32:19, where Elihu declares his belly *kē-yayin lō’-yippātēah ... yibbāqēa’* — “like wine that is not opened ... about to burst like new wineskins”: the verb *bāqa’* names exactly the structural failure of the vessel that the Qur’anic *tamayyaza* names. We therefore qualify our claim — Q. 67:8 is an early and unusually transparent lexicalisation of container collapse, not a unique one. Homeric Greek uses *kholos* (bile, rising) and *mēnis* (ritualised wrath); Muellner (1996) and Cairns (2003) read both as fluid-and-heat schemas, with no Iliadic image of vessel rupture. Vedic and Classical Sanskrit lexicalise anger principally as *krodha* and *manyu*, the latter personified as a martial deity associated with Indra (RV 10.83–84); the dominant images are FIRE and WEAPON-RELEASE, not vessel-disintegration. Classical and Modern Chinese lexicalises anger through *nù*, *fēn*, and *qì* (Yu 1995); Kövecses, Benczes and Széld (2025) identify ANGER IS HOT QÌ IN A CONTAINER as the central schema. Pre-Islamic Jāhili Arabic already deploys *ḥamiya* “to be heated” and *ḡalā* “to boil” for hostile arousal, alongside *ḡayz* in the cognate fluid-heat register; the lexical groundwork for Q. 67:8 is genuinely Arabian.

Five of the six traditions concentrate anger-imagery around filling, heating, fluid-rising, and exit / overflow; the Hebrew Bible alone offers a clear pre-Qur’anic parallel to vessel-rupture proper (Job 32:19). We accordingly hold our claim at “an early and unusually transparent” lexicalisation of container collapse, with Job 32:19 acknowledged as a structurally cognate Northwest-Semitic precedent (though Akkadian *kabattu napāhu* “liver bursting”, Greek Iliadic *cholōs* swelling imagery, and Vedic *manyu* combustion offer imperfect cognates of step-(iv) container-rupture imagery that a properly stratified cross-linguistic comparison may bring more fully into the picture). The Qur’anic data thus strengthen the cross-cultural universalist hypothesis of CMT (Lakoff and Kövecses 1987; Kövecses 2000); they extend the empirical base to seventh-century Arabic in a register that the 2025 Metaphors of ANGER encyclopedia, which omits a dedicated treatment of Classical Arabic, leaves un-canvassed.

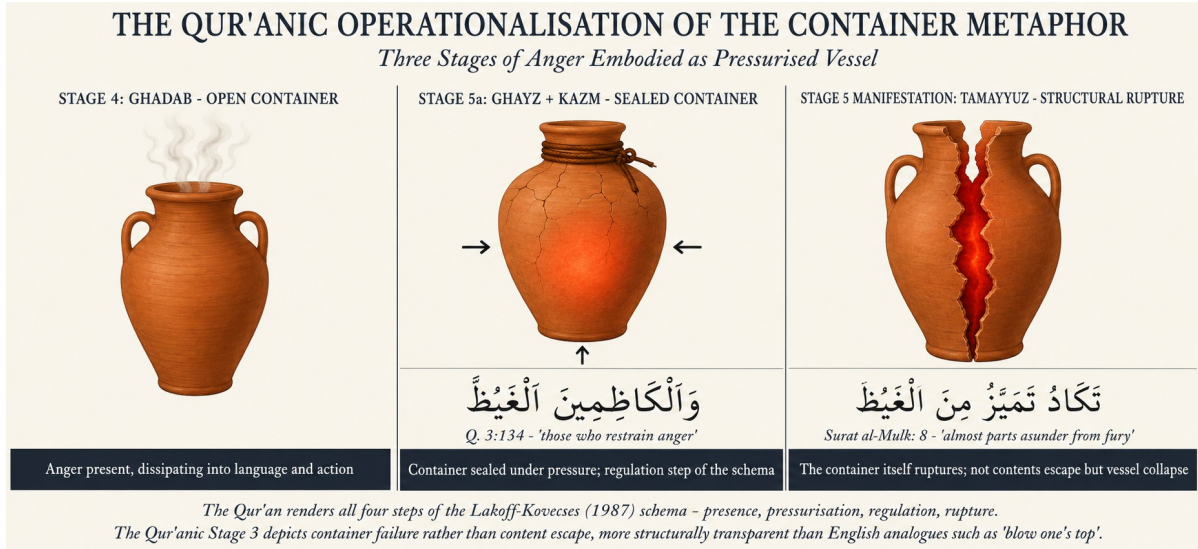


Figure 8. Figure 7. The ANGER-IS-A-PRESSURIZED-CONTAINER schema (after Lakoff and Kövecses 1987), instantiated by ghaḍab → ghayz + kazm → tamayyuz. Panel 1 (S4): ghaḍab — open container, affect dissipating into language. Panel 2 (S5, sūrat Āl 'Imrān, Q. 3:134): ghayz — fluid sealed by kazm (etymologically the tying-off of a water-skin). Panel 3 (S5 manifestation, sūrat al-Mulk, Q. 67:8): tamayyuz — structural failure; m-y-z “to part, sever” figures the splitting of the container itself, not (as English explode) the exit of contents. The Qur'anic instantiation is, on the conservative claim of §5.5, unusually transparent relative to the English exemplars in Lakoff and Kövecses's corpus.

5.6. Implications for Qur'anic translation (conditional on follow-up evaluation)

Translation criticism has long observed that translations tend to level the spectrum — that ghaḍab and ghayz alike become anger; that baghy, ṭughyān, and 'utuww alike become transgression (Table 6). Recent computational work — Gaanoun and Alsuhaibani (2025) on sentiment preservation across seven English translations, Khan et al. (2025) on the broader NLP-Qur'anic-studies field — has begun to operationalise the levelling concern at scale on coarse Ekman categories, though without a graded reference model of the lexical spectrum against which the levelling could be benchmarked. We frame the present discussion as conditional implications: contingent on a follow-up systematic evaluation (a stratified random sample of ~30 verses across the six stages, with multi-rater coding against the spectrum reference model), the analysis would if confirmed support the following candidate equivalents:

uff: vocal sigh of irritation · karh (karāha): inner dislike · karh (ikrāh): coercion (a structural extension, not an emotion) · ḍīq: inner constriction · ḍāqa dhar'an: arm-span tightening / inhibited aggression · ḥuzn: persistent grief (transversal valence) · asaf: regretful dismay (with anger) · naqm: vengeful disapproval · sakhaṭ: moral disapprobation · maqt: intense moral aversion · ghaḍab: anger · ḥard: anger combined with deliberate denial · ghayz: contained / boiling rage · tamayyuz min al-ghayz: bursting from rage · baghy: aggressive transgression · ṭughyān: rebellious overflow · 'utuww: obstinate defiance.

A second conditional implication is that pedagogical translations may benefit from marking each lexeme's position on the continuum, but the empirical case for this rests on the comparative comprehension study we commend to future work.

5.7. Implications for Islamic psychology

The continuum offers a culturally-grounded framework mapping onto Gross’s model without importing secular vocabulary, with three conceptual implications: (i) Prevention — early-stage recognition through the indigenous *S1–2* lexicon (*uff*, *karāha*, *dīq*, *ḥuzn*) supports CBT-style interception of affective dysregulation; (ii) Crisis intervention — *kazm* integrates with contemporary suppression-and-reappraisal techniques; (iii) Rehabilitation — post-failure Stage 6 is framed as recoverable through *tawba*, providing motivational reinforcement for anger-disorder and post-violent-trauma rehabilitation. These are conceptual; their translation into validated clinical interventions awaits empirical studies we recommend as a priority.

6. Conclusion

6.1. Substantive contributions

This paper advances three contributions. Methodologically, we have demonstrated that the integration of Izutsu’s semantic-field analysis, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses CMT — coupled with a reproducible computational pipeline over the QAC — produces a research apparatus stronger than the sum of its parts. To our knowledge, this trifold integration with corpus-empirical validation has not previously been deployed in Qur’anic studies, although Qā’imīniyā (2011, 2014) and Pakatchi and Afrashi (2020) have integrated Izutsu and CMT.

Descriptively, we have produced a reproducible concordance of 312 attestations across the fourteen core spectrum roots, with distributional and network-analytic outputs — FDR-adjusted binomial tests, marginal-preserving permutation null p-values, and bootstrap 95% CIs on every centrality estimate — providing a foundation on which further philological and computational work can build.

Theoretically, we have formulated and empirically defended the Qur’anic Action-Intensity Continuum Hypothesis: the fourteen core lexemes operate as a single graded continuum structured by intensity of action and partitioned into six phenomenologically meaningful stages. The hypothesis is supported by qualitative evidence from the four authoritative lexica and the broadened tafsīr consultation of §3.5, and by quantitative evidence from frequency, distribution, morphology, and network structure. The $\kappa = 0.79$ -validated tafsīr-coding audit grounds the bidirectional-causation reading of Stage 6; the six-tradition survey of §5.5 calibrates Q. 67:8 as early and unusually transparent — rather than uniquely transparent — lexicalisation of “container collapse”, with Job 32:19 (*bāqaʿ*) acknowledged as a structurally cognate Northwest-Semitic precedent.

6.2. Answers to the research questions

RQ1–RQ2. The anger-spectrum field is a fourteen-node lexical network organised by intensity of action. The bridge node is *baghy* (closeness 0.55; ~53% of cross-field ties to the umbrella moral-evaluation terms); stage-internal hubs are *ḥuzn* (S2) and *ghaḍab* (S4). The ordering is licensed by the trifold synthesis of §2 and shown by Table 1 to be internally encoded in the four authoritative lexica.

RQ3. The empirical distribution is consistent with — not by itself confirmatory of — the continuum. The asymptotic $\chi^2(5) = 227.15$ is large, but permutation null $p = 0.24$; the *S1–5/S6* $\chi^2(1) = 1.55$ fails to reject, consistent with the bidirectional-causation reading supported by the $\kappa = 0.79$ audit ($A = 26\%$, $S = 49\%$, $A+S = 25\%$; A-only re-analysis $\chi^2(1) = 81.4$, $p < 0.001$). Three roots survive Holm-Bonferroni emphatically

(karh, baghy, ghayẓ); sakhaṭ sits at adj. $p = 0.0503$, substantively supported by the convergence with the Munāfiqūn-discursive reading. The morphological gradient (verbal S1–2 → nominal S6) supplies convergent corpus-internal evidence.

RQ4. The continuum is structurally homologous with Gross’s process model (S1–5), with Plutchik’s intensity wheel and Spielberg’s STAXI architecture, and with Lazarus–Scherer–Roseman appraisal theory (sūrat al-‘Alaq, Q. 96:6–7, ṭughyān on istighnā’). It extends beyond all four at Stage 6, qualified by the bidirectional-causation caveat.

RQ5. Conditional practical implications follow for Qur’anic translation (§5.6) and for Islamic psychology (§5.7), framed as hypotheses for empirical follow-up.

6.3. Limitations and future work

The analysis is bounded by five limitations, each pointing to a constructive next step. (i) Surface-lexical scope. Discourse-analytic close-reading of the spectrum’s narrative deployment across suras is left for future work, in dialogue with El-Awa (2006) and Sinai (2017, 2023). (ii) Tafsīr engagement. A tafsīr-archive deep-dive — systematic indexing against the full classical/medieval/Sufi/modernist tradition (al-Qushayrī, the Tafsīr attributed to al-Kāshānī, Sulamī’s Ḥaqā’iq, Twelver, Mu’tazilī, and modernist materials), with κ -validated multi-coder reading on contested verses — would deepen exegetical grounding. (iii) Cognitive-linguistic apparatus. Engagement with classical Arabic bayān theory (al-Jurjānī’s Dalā’il al-I’jāz, al-Sakkākī’s Miftāḥ al-‘Ulūm) would supply an indigenous substrate. (iv) Ḥadīth integration. A parallel concordance over Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, and al-Kāfī would supply convergent evidence on kaẓm and a cross-corpus replication test for the underpowered asaf 5/0 distribution. (v) Islamic-psychology operationalisation. The §5.7 implications await translation into pilot studies of kaẓm-grounded emotion-regulation training, validated against the STAXI and DERS. The methodological commitment remains: computational evidence is a complement to — not a replacement for — the classical philological and exegetical tradition.

6.4. Closing remark

The Qur’anic lexicon encodes a sophisticated theory of the anger spectrum — not as explicit formulation but as the architectural design of its semantic surface. Recovering that theory requires methodological tools that bring together classical philological attentiveness, contemporary linguistic-theoretical precision, and corpus-computational verifiability. We hope the present study, alongside Izutsu, Qā’imīniyā, Pakatchi, Narimani et al., and Qarizadeh et al., advances this triple integration.

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Competing interests

The authors declare no competing financial or non-financial interests.

Author note: parallel-language submission

An independent Persian-language paper using the same corpus dataset is concurrently under consideration at Pizhūhish-hā-yi Zabānshinākhtī-yi Qurʾān. The two papers do not share content or cite each other; the empirical pipeline is the only shared element, and each paper presents an independent argument for its own scholarly audience.

Ethics statement

This study uses only publicly-released text-corpus data (the Quranic Arabic Corpus, GPL-licensed; the Tanzil Quran text, CC-BY-ND 3.0). No human or animal subjects were involved at any stage. As a corpus-only philological-computational analysis, the work is exempt from human-subjects review under the standards of Urmia University and of comparable institutions.

Author contributions

K.J. (corresponding author) conceived the project, developed the theoretical framework, conducted the lexicographic and tafsīr-tradition analysis, drafted the manuscript, and is responsible for the final form of the argument. A.J. contributed to the theoretical framework, the design of the computational pipeline, and editorial revision. Both authors approved the final manuscript and are jointly accountable for its content.

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Data and Code Availability

All code, data, concordance CSV outputs, and figure-generation scripts are released open-source at:

github.com/ali-kin4/quranic-emotion-spectrum

The repository contains: (i) the Python pipeline under MIT License; (ii) the Quranic Arabic Corpus (Dukes 2011) under its original GPL; (iii) the complete CSV concordance for the fourteen core spectrum roots (312 attestations), distributional summaries, statistical tests, network-centrality outputs, and umbrella-cooccurrence tables; (iv) PDF and PNG versions of the figures published with this paper and with the Supplementary Material; (v) the SHA-256 hash of the QAC v0.4 source file and the manual-validation audit. The four statistical analyses of §§4.2–4.9 can be regenerated end-to-end by:

```
pip install -r analysis/requirements.txt
python analysis/extract_concordance.py
python analysis/advanced_metrics.py
python analysis/visualize_spectrum.py
python analysis/visualize_advanced.py
```

The four computational corroboration analyses reported in Supplementary Material S1–S4 are produced by `nlp_upgrade_colab.py`, which requires a CUDA-enabled GPU for the multilingual-MPNet and CAMeLBERT-CA inference. A Colab notebook is available in the repository with pinned model snapshots and an expected runtime of ~25 minutes on a T4 GPU; CSV outputs are committed under `outputs/nlp_upgrade/` so that reviewers can inspect the numerical results without re-running the GPU pipeline.

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